
GenoLeWM: An Action-Conditioned Latent World Model for Genomic Edits.

Post-Release Validity Correction for v0.2.1-r1

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<https://github.com/AbdelStark/GenoLeWM>

Abstract

GenoLeWM is an action-conditioned latent-prediction pipeline that freezes Carbon-500M and trains a cross-attention predictor over explicit genomic edits. This paper originally interpreted the released v0.2.1-r1 benchmark as a negative scientific result. A post-release audit invalidates that interpretation. Although the release configuration set `normalize: true`, the encoder returned raw pooled source and target states, while the predictor always returned a unit-norm vector. Training also used incompatible pooling coordinates: reference states fell back to global-mean pooling because no edit locus was supplied, whereas targets were pooled around the edit; rollout source and target states were edit-centered, and historical distractors could be centered elsewhere. Moreover, every historical “centered” pool omitted the leading `<dna>` control-token offset and was shifted one hidden token left. The pinned Carbon tokenizer code also made an unpinned, network-capable transitive load of Qwen/Qwen3-4B-Base, so its local hash did not define a self-contained runtime. On the released rollout slices, mean source/target norms were 33.982/33.595 (phased) and 29.253/29.089 (synthetic), whereas mean prediction norm was 1.000 on both. Consequently, released L_2 losses, surprise scores, L_2 rollout metrics, and conclusions derived from them do not evaluate the intended normalized, coordinate-matched method. The released cosine values (0.289 and 0.302, versus source-state values 0.998 and 0.991) remain reproducible historical implementation measurements because cosine is scale-invariant, but they are confounded by invalid mixed-scale, mixed-pooling training and a train/rollout distribution mismatch; they do not establish model capability or a failure mechanism. The actual released run used seed 271828, 10,000 optimizer steps and 80,000 samples. It was labeled `phase2`, but used no LoRA, and its target-only Gaussian KL term was constant with respect to trainable parameters and supplied no gradient. We preserve the released numbers for auditability, withdraw the mechanistic attribution and capability conclusions, and require a corrected end-to-end run under a new content-addressed identity before making either positive or negative scientific claims.

Post-Release Validity Correction

Correction issued 10 July 2026. This version supersedes the original interpretation of all v0.2.1-r1 model-quality results. The release receipts establish included-artifact identity, not semantic or transitive-runtime validity, and therefore did not detect that `normalize: true` was ignored, that pooling omitted the leading DNA control-token offset, or that the pinned tokenizer code could fetch an unpinned Qwen tokenizer. Four boundaries now govern the paper: (i) all reported values remain available only as historical measurements of the released implementation; (ii) all L_2 -dependent results, including surprise-based VEP, are invalid as evaluations of the intended method;

(iii) cosine measurements are retained but cannot support a capability or causal conclusion because training used mismatched scale and pooling coordinates while rollout evaluation used a different distribution; and (iv) no corrected result may reuse the v0.2.1-r1 identity. A new checkpoint, dataset/evaluation manifest, metric report, and receipt graph are required after normalization, token-layout, pooling-coordinate, and self-contained-runtime contracts are fixed and tested end to end.

1 Introduction

DNA foundation models [8, 12, 15, 16, 19] answer a representational question: *what is the embedding, or the likelihood, of this sequence?* A different and largely unanswered question is *interventional*: given a reference genomic window and an explicit edit (a single-nucleotide variant, an insertion, or a deletion), can a model predict the *consequence* of that edit directly in representation space, cheaply enough to score variants, roll out multi-edit haplotypes, and plan edit sets? An affirmative answer would turn a static genomic encoder into an action-conditioned *world model* [9, 10] whose actions are discrete genomic edits.

GenoLeWM tests this question in its most economical form. The intended method freezes a pre-trained DNA encoder, Carbon-500M [11], and trains only a small predictor: a reference window w_{ref} is encoded at the edit locus to a state $s_t = \text{enc}(w_{\text{ref}}) \in \mathbb{R}^{1024}$, an edit action a is embedded by a lightweight action encoder, and a cross-attention predictor estimates the post-edit state $\hat{s}_{t+1} = g(s_t, a)$, supervised against a coordinate-matched frozen encoding $s_{t+1} = \text{enc}(w_{\text{alt}})$ of the edited window (fig. 1). The implementation also exposes surprise scoring, autoregressive rollout, and CEM planning. The released experiment, however, did not instantiate the intended state contract: encoder states were raw, training source and target pooling differed, and predictor outputs were unit-normalized. We therefore describe the architecture but do not treat v0.2.1-r1 as a valid test of it.

The headline is a validity correction. The released benchmark cannot determine whether GenoLeWM beats, matches, or trails its intended baselines. Historical VEP deltas include -0.1875 (ClinVar coding AUROC) and -0.3125 (ClinVar non-coding AUROC), while historical rollout cosine is ≈ 0.29 versus ≈ 0.99 for the source-state baseline. Those values describe the flawed implementation, not the normalized-state method. In particular, the VEP score and rollout L_2 metric compare unit-norm predictions with targets whose mean norms are about 29–34. The content-addressed package makes this failure reproducible, but its receipts bind identities and bytes; they do not validate the semantics of the computation.

Contributions after correction. This paper now makes four narrower contributions.

1. **An implemented research architecture** (section 3): a discrete-edit action encoder, cross-attention predictor, rollout wrapper, surprise interface, and CEM planner. No behavioral capability is inferred from the released run.
2. **A content-addressed train→eval→benchmark→replay pipeline** (section 4) whose strengths and limits are now explicit: it binds artifacts and outputs, but does not establish semantic correctness.
3. **A post-release validity audit** (sections 5 and 6) that preserves the historical measurements, reports the state-norm and token-layout mismatches, discloses the unpinned transitive tokenizer, corrects the run metadata, and identifies which metrics and conclusions must be withdrawn.
4. **A falsifiable correction protocol** (section 10) requiring self-contained tokenization, token-layout-derived centers, consistent state normalization, gradient-bearing regularization if claimed, distribution-matched evaluation, and a new content-addressed run identity before scientific interpretation resumes.

What we do not claim. GenoLeWM is not a clinical, diagnostic, or decision-support tool. We make no model-quality, world-model-capability, surprise-validity, planning-usefulness, or mechanistic claim from v0.2.1-r1; we also make no privacy, runtime-execution, or efficiency guarantee. The released VEP, rollout L_2 , and planning-distance results are not valid evaluations of the intended normalized-state method. The historical cosine results are insufficient for interpretation rather than evidence for a specific failure mode. We collect these boundaries in section 7.

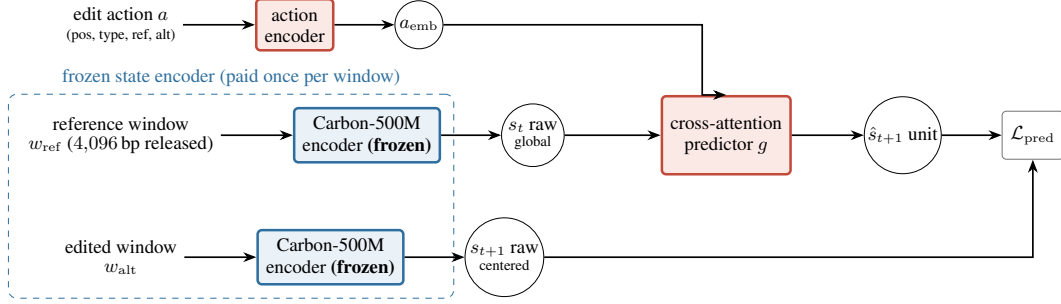


Figure 1: **Released v0.2.1-r1 computation, exposing the state-contract violation.** A frozen Carbon-500M DNA encoder maps a reference window w_{ref} to a state $s_t \in \mathbb{R}^{1024}$; a small trainable action encoder maps an explicit edit $a = (\text{pos}, \text{type}, \text{ref}, \text{alt})$ to a_{emb} ; and a cross-attention predictor g returns $\hat{s}_{t+1} = g(s_t, a_{\text{emb}})$. Although `normalize: true` was configured, released source and target encoder states were raw (rollout mean norms 29–34), while the predictor output was unit norm. Training sources used global pooling while targets used edit-centered pooling; every nominal edit center also omitted the leading `<dna>` token and was shifted one hidden token left. Rollout changed the source coordinate again. The frozen target-only KL had no gradient and no LoRA was present. The diagram documents implementation flow, not validated world-model capability.

2 Related Work

GenoLeWM sits at the intersection of four research programs. DNA foundation models supply the encoder; variant-effect prediction supplies the evaluation and the foil; joint-embedding predictive architectures supply the intended objective; and latent world models supply the action-conditioned framing and planning interface. The released experiment does not support a scientific comparison to these programs because it violated its state contract.

DNA foundation models and genomic functional prediction. Masked, causal, and state-space sequence models include DNABERT [12], the multi-species Nucleotide Transformer [8], HyenaDNA [15], Caduceus [19], and the genome-scale generative models Evo and Evo 2 [5, 16]; they expose sequence representations or likelihoods. AlphaGenome [2] instead predicts functional genomic tracks and can compare reference and alternate sequences. Carbon-500M [11] is the frozen sequence encoder used here. The intended GenoLeWM distinction is narrower: a structured edit action conditions a learned latent transition operator. The released experiment does not establish that this operator is valid.

Variant-effect prediction and its resources, and the precise sense in which we are not a clinical predictor. The released evaluation used ClinVar clinical interpretations [14], gnomAD population frequencies [13], TraitGym [4], the Sahu et al. BRCA2 saturation genome-editing functional assay [18], and a Carbon-500M zero-shot baseline; AlphaMissense [6] provides broader field context. GenoLeWM is not a clinical predictor. No pathogenicity label enters predictor training, and no output should be treated as a decision. More importantly after this correction, the released surprise score was an L_2 residual between unit-norm predictions and unnormalized target states. Its ranking metrics therefore cannot answer even the narrow internal question of whether action conditioning adds signal over Carbon-500M. We retain the benchmark rows as provenance records, not clinical or comparative evidence.

JEPAs and anti-collapse SSL: intended objective versus released computation. The objective lineage is the Joint-Embedding Predictive Architecture. I-JEPA [1] predicts in representation space, and LeJEPA [3] motivates distribution regularization. GenoLeWM intended to combine cosine and normalized-state L_2 prediction terms and, in Phase 2, a Gaussian regularizer with encoder adaptation. The released run did something materially different: its configuration said `phase2`, but Carbon-500M remained frozen with no LoRA; the KL was computed only from frozen target states, making it constant with respect to trainable parameters; and the L_2 term compared unit-norm predictions with raw target states. Thus neither the intended normalized prediction objective nor a gradient-

Table 1: Architectural positioning. The released GenoLeWM run used a frozen encoder and discrete genomic-edit actions, but its normalization and regularization mismatches prevent behavioral comparison.

Method family	Learns encoder?	Anti-collapse	Action	Planning
DNA FMs [8, 12, 15, 16]	yes (pretrain)	n/a	none	none
Supervised VEP [6]	task-tuned	n/a	none	none
JEPA / LeJEPA [1, 3]	yes, jointly	EMA / Gaussian reg.	n/a	none
Latent world models [7, 9, 10]	yes	recon./reward	continuous / discrete control	imagination / MPC
GenoLeWM (released)	no (frozen)	target KL (no gradient)	discrete edit	CEM interface

bearing Phase-2 regularizer was evaluated. Any proposed latent-residual or representation-geometry explanation remains a hypothesis for a corrected experiment, not a finding of this release.

Latent world models and planning: a world model whose action is a discrete genomic edit.

The architectural lineage runs from Ha and Schmidhuber’s World Models [9] through the Dreamer family [10] to sampling-based latent model-predictive control. These systems may support continuous or discrete control actions; GenoLeWM’s distinction is the structured genomic semantics (position, type, ref, alt) and a factored CEM search space. That interface is an architectural research scaffold. Because both training and latent-distance evaluation were affected by the state-contract violation, the released planning trace demonstrates code-path execution only; it does not show a valid transition model or planning capability.

Table 1 summarizes the positioning.

3 Method

GenoLeWM has three trained-or-frozen components (a frozen state encoder, a trainable action encoder, and a trainable predictor) and three downstream operations: surprise scoring, rollout, and planning. We distinguish the intended normalized-state method from the released v0.2.1-r1 computation. This distinction is load-bearing: configuration intent is not evidence of runtime behavior.

3.1 State encoder (frozen)

The encoder is Carbon-500M, a 500M-parameter decoder-only DNA language model (LLaMA-style, hidden size 1024), loaded at a pinned revision in bf16 and frozen in the released run. The code default is a 12,288 bp window, but the released v0.2.1 placed-window manifest supplied 4,096 bp windows (padded to the tokenizer’s six-base multiple). The intended representation pools layer-20 hidden states by a centered mean over a ± 8 -token window (± 48 bp) around one shared edit locus. Carbon emits one leading `<dna>` control token, so let

$$u_t = \text{mean}\left(H_L[c-r : c+r+1]\right), \quad s_t^{\text{intended}} = \frac{u_t}{\|u_t\|_2}, \quad c = 1 + \lfloor \text{locus}/6 \rfloor, \quad r = 8. \quad (1)$$

The configuration set `normalize: true`, but the released encoder returned raw pooled vectors without applying the second operation. Worse, training called the source encoder without an edit locus, which triggered global-mean fallback, while targets used a historical “centered” operator with $c_{\text{hist}} = \lfloor \text{locus}/6 \rfloor$. It omitted the leading control token, shifting every intended center one hidden token left and sometimes centering the pool on `<dna>`. Rollout generation used the same shifted rule for source and target states at the first target edit, but centered historical distractors at each distractor’s own first edit. Thus the release mixed scale, coordinate frame, and token layout. The reported norm audit concerns these shifted rollout rows: 33.982/33.595 for source/target phased states and 29.253/29.089 for source/target synthetic states. These were not benign metadata discrepancies: they changed optimization, retrieval geometry, and every L_2 -based score. Cache v1 also omitted the pooling center from its key, so centered states for the same sequence could collide. The ± 48 bp radius is additionally narrower than the RFC’s ± 1536 bp default.

The pinned Carbon tokenizer.py created a separate reproducibility failure: it called Qwen/Qwen3-4B-Base without a revision or local-only constraint. Pinning and hashing the local Carbon custom-tokenizer file therefore did not bind the transitive tokenizer implementation and did

not make the runtime self-contained. The corrected source path includes its local pure-DNA tokenizer implementation in encoder identity, validates the control-token/six-mer layout, and derives c from the observed DNA-content start. These are code-contract repairs only; no corrected checkpoint or model-quality result follows from them.

3.2 Action encoder (trainable)

An edit is represented as an `EditSpec` (absolute VCF coordinates chrom, pos, ref, alt) or its window-relative form `RelEdit`, with edit type derived from the (ref, alt) lengths into {SNV, INS, DEL, MNV, INDEL} and a 16 bp cap on either allele. The action encoder maps an edit to $a_{\text{emb}} \in \mathbb{R}^{64}$ through four sub-encoders, namely a base-pair-resolution sinusoidal position embedding ($\mathbb{R}^{128}$), a learned type embedding ($\mathbb{R}^{64}$), and a *shared* small Transformer (“SeqMicroEncoder”) applied to the reference and alternate alleles ($\mathbb{R}^{256}$ each); these are concatenated to $\mathbb{R}^{704}$ and projected by a two-layer MLP with GELU and LayerNorm. The sinusoidal schedule uses the window length as its frequency base rather than the conventional 10,000.

3.3 Predictor (trainable, residual)

The predictor g is a cross-attention Transformer over $[s_t; a_1; \dots; a_K]$, with six cross-attention blocks, two self-attention blocks, hidden dimension 768, eight heads, GELU, pre-LayerNorm, learned token-type and step-position embeddings, and a causal mask. Its residual read-out is always L_2 -normalized, with the final MLP layer zero-initialized:

$$\hat{s}_{t+1} = \frac{s_t + \text{MLP}(g(s_t, a_{\text{emb}}))}{\|s_t + \text{MLP}(g(s_t, a_{\text{emb}}))\|_2}. \quad (2)$$

Under the intended unit-norm contract, zero initialization would make $\hat{s}_{t+1} = s_t$. Under the released contract it instead made $\hat{s}_{t+1} = u_t / \|u_t\|_2$, while the reported source-state baseline remained the raw u_t . Thus the previous claim that initialization exactly equaled the evaluated “predict-no-change” baseline was false for this run. An autoregressive wrapper (ARPredictor) unrolls the predictor with KV-caching and upcasts the output MLP to fp32 beyond horizon 20.

3.4 Released training objective (phase2 label, frozen encoder)

The prediction term was

$$\mathcal{L}_{\text{pred}}(\hat{s}_{t+1}, s_{t+1}) = \alpha (1 - \cos(\hat{s}_{t+1}, s_{t+1})) + \beta \frac{\|\hat{s}_{t+1} - s_{t+1}\|_2^2}{d}, \quad \alpha=1.0, \beta=0.1, d=1024, \quad (3)$$

averaged uniformly over rollout steps with a mask. Because $\hat{s}_{t+1}$ had norm 1 and s_{t+1} had norm about 29–34, the L_2 term did not measure the intended unit-sphere residual. The run was configured as `phase2`, so the implementation added $\gamma \mathcal{L}_{\text{reg}}$ with $\gamma=0.5$, where $\mathcal{L}_{\text{reg}} = \frac{1}{2}(\|\mu\|^2 + \text{tr } \Sigma - \log \det(\Sigma + \epsilon I) - d)$. However, $\mathcal{L}_{\text{reg}}$ was computed only from frozen target states. No LoRA parameters existed, and the term was constant with respect to the predictor and action encoder; it supplied no optimization gradient. The released run used AdamW, seed 271828, 10,000 optimizer steps, 80,000 samples, and deterministic mode. The previously reported seed 104729 and 20,000 steps describe a different first-experiment lineage, not `v0.2.1-r1`.

3.5 Surprise scoring

The intended raw surprise is the normalized-state residual $\sigma_{\text{raw}}(v) = \|g(s_t, a_v) - \text{enc}(\text{apply}(v, w_{\text{ref}}))\|_2$, calibrated by an empirical context-bucket CDF. In the release, the first operand was unit norm and the second was raw. Calibration can rank a flawed quantity consistently but cannot restore its intended meaning. Accordingly, all released surprise-based VEP metrics are historical implementation measurements and are withdrawn as evidence about variant effects or action conditioning.

3.6 Latent planning

Given an initial state and a target latent, the CEM planner samples discrete edit sequences, scores predictor rollouts by latent distance, and refits a factored proposal. It uses defaults of horizon 5, five

Table 2: **Released computation versus intended or previously reported behavior.** The released column records what produced the historical measurements; it is not a reproduction recipe for the intended method. A valid rerun must use a new identity.

Item	Released (v0.2.1-r1)	Intended / previously reported
Encoder pooling layer L	20	final (-1)
Pool radius r	8 tok (± 48 bp)	256 tok (± 1536 bp)
Training source pooling	global mean fallback	centered at edit locus
Training target pooling	nominal center; one token left	same validated center as source
Rollout candidate pooling	candidate-specific	one shared comparison center
Center token index	$[\text{locus}/6]$	$1 + \lfloor \text{locus}/6 \rfloor$ after <code><dna></code>
Tokenizer dependency	unpinned network-capable Qwen load	self-contained local pure-DNA path
Placed-window length	4,096 bp	12,288 bp default / previously reported
Encoder state normalization	raw; flag ignored	unit norm (<code>normalize: true</code>)
Predictor output norm	1.000	unit norm
Source/target mean norm (phased)	33.982/33.595	1/1
Source/target mean norm (synthetic)	29.253/29.089	1/1
State dim d_{state}	1024	1024 (package default 512, overridden)
Action dim d_{action}	64	512
Predictor cross + self blocks	6 + 2	4 + 2
Predictor hidden dim d_{hidden}	768	1024
FFN intermediate dim	768 (i.e., $1 \times$)	2048 ($2 \times$)
Warmup steps	1000	2000
Weight decay	0.1	0.05
Physical batch size	8	256 effective
Run label / adaptation	phase2; no LoRA	Phase 2 with trainable LoRA
Gaussian KL	target-only constant; no gradient	gradient-bearing regularizer
Seed / optimizer steps / samples	271828 / 10,000 / 80,000	104729 / 20,000 previously reported
Edit-source mix (gnomAD/synSNV/synINDEL/ClinVar)	3:3:1:1	4:3:2:1

iterations, 1024 samples, and 64 elites. The search path executes without further encoder calls after setup, but its released distances inherited the incompatible state scales. The trace is therefore integration evidence only, not evidence that the planner optimized a scientifically meaningful objective.

3.7 The efficiency thesis and its honest scope

The design rationale is amortization: pay one Carbon-500M forward pass per reference window, then issue latent queries through the smaller predictor. Rollout and planning make no further encoder calls after setup, whereas single-variant surprise scoring requires an edited-window encode. These are call-graph properties, not demonstrated efficiency or utility. The released latency benchmark exercises a cold subprocess and the model-quality path is invalid, so no end-to-end efficiency–quality claim follows.

4 Systems and Reproducibility

The most defensible positive contribution of this work is infrastructural but narrower than originally stated: the released artifacts and numbers are content-addressed and replayable. The validity correction demonstrates what this does *not* provide. Reproducible execution can faithfully reproduce a semantically invalid computation, so provenance must be paired with tested scientific contracts [17].

Content-addressed provenance. A canonical-JSON serialization (sorted keys, rejected NaN/Inf, UTF-8) feeds a SHA-256 scheme in which the model identity hashes the model manifest and its artifact files. Each scoring call emits a checksum receipt binding the model id, input commitment, output, calibration hash, and runtime metadata. These receipts certify byte-level integrity and identity for included files, not attestation or scientific validity. In particular, the input commitment recorded `normalize: true` while runtime behavior ignored it. Historical encoder identity also omitted the runtime package, and even hashing the later local `tokenizer.py` would not have bound its unpinned transitive Qwen load. The receipt therefore committed to included bytes and configuration intent without proving the state transformation or a self-contained runtime.

Evidence-bound benchmark suite. The benchmark suite executes a fixed step graph and records the path, SHA-256, and size of each output. The release reported 28 verified outputs and all required benchmark rows. That proves coverage of the declared graph, not correctness of the state representation or metric operands. The normalization contract was absent from the semantic gate.

Why the original verifier was insufficient. The paper toolchain re-renders tables from artifacts and rejects stale text, placeholders, and mutated evidence. It also required selected negative deltas to be present. Those checks prevented transcription drift but mistook sign constraints for validity constraints: they never checked source, target, and prediction norms; whether `normalize: true` affected runtime states; whether base-pair loci mapped after the DNA control token; whether tokenizer dependencies were local and pinned; whether a claimed regularizer had a gradient path; or whether train and rollout distributions matched. A corrected release gate must establish those invariants before accepting any metric sign, and the corrected run must receive a new identity rather than overwriting this historical graph.

On-device posture. The current source runtime enforces a fail-closed network guard during inference and uses a self-contained pure-DNA tokenizer; the released lineage did not satisfy that tokenizer boundary because its custom code could perform the unpinned Qwen lookup during setup. The project also ships local-first importers for consumer personal-genome formats, with a redaction filter that drops DNA-like strings and a deny-list of personal fields at the logging boundary. We describe these as current engineering choices for local execution; we explicitly do *not* present them as historical runtime proof or a privacy guarantee.

5 Experiments

Setup. The historical measurements come from one released identity (table 6): model `geno-lewm-v0.2.1-r1`, dataset snapshot `geno-lewm-data-v0.2.1-r1`, commit `d9b06815`, evaluated on one NVIDIA H200. The checkpoint used seed 271828, 10,000 optimizer steps, and 80,000 samples. Its configuration said `phase2`, but Carbon-500M was frozen, no LoRA was present, and the target-only KL supplied no gradient. The historical baselines were Carbon-500M zero-shot for VEP and an unchanged raw source state for rollout. We report the rows to document the invalidated release, not to compare scientific methods.

5.1 Variant-effect prediction

Table 3 and fig. 2 preserve the historical VEP rows. The GenoLeWM score was derived from the invalid L_2 residual and therefore is not a measurement of the intended surprise function. The historical values include ClinVar coding AUROC 0.734 ($\Delta = -0.1875$), ClinVar non-coding AUROC 0.563 ($\Delta = -0.3125$), BRCA2 Spearman 0.149 ($\Delta = -0.328$), and TraitGym Spearman -0.028 ($\Delta = +0.056$). These numbers remain useful for tracing the released artifact but support neither superiority nor inferiority. Independent of the normalization bug, the slices are also small ($N=16$ for each ClinVar slice, $N=32$ for the BRCA2 continuous-score slice, and $N=32$ for the TraitGym binary-label slice), so a corrected rerun must be powered before interpretation.

5.2 Latent rollout fidelity

The rollout track (table 4, fig. 3) exposes the state-contract violation. Mean source/target/prediction norms are 33.982/33.595/1.000 for phased haplotypes and 29.253/29.089/1.000 for synthetic edit chains. Hence the historical L_2 values and deltas are invalid comparisons. Historical cosine remains numerically reproducible and scale-invariant: prediction-to-target cosine is 0.289 and 0.302, while raw-source-to-target cosine is 0.998 and 0.991. But the predictor was trained from global-mean source states toward targets whose intended centers were shifted one hidden token left under the invalid mixed-scale objective, then evaluated from rollout sources using the same shifted centering rule; candidate ranking also mixed loci. These cosine values cannot isolate normalization, token layout, pooling coordinates, optimization, distribution shift, action use, or encoder geometry. Recall@ k is 1.0 for both predictor and baseline at $N=8$ and is non-discriminating.

5.3 Inference efficiency

Table 5 and fig. 4 report a cold `geno-lewm-score` subprocess measurement: median latency 115,262.94ms, throughput 0.31 variants/s, and peak memory 1.83 GiB. It includes interpreter startup, model loading, and two Carbon-500M forward passes, but no component profile was recorded. Warm-cache and latent-only regimes were not measured and are shown without

Table 3: **Historical variant-effect measurements (v0.2.1-r1; invalid for the intended method).** Binary ClinVar slices report accuracy / AUROC / average precision (AP) / balanced accuracy; BRCA2 saturation and TraitGym Mendelian report Spearman ρ against a BRCA2 continuous functional score and TraitGym binary labels. The baseline is the same frozen Carbon-500M encoder scored zero-shot. Δ is GenoLeWM – Carbon (positive favours GenoLeWM). The GenoLeWM score was derived from an L_2 residual between unit-norm predictions and raw target states whose norms were about 29–34. Thus the rows preserve released bytes but do not compare the intended normalized-state method with Carbon. Bold marks only the larger historical value.

Benchmark	Metric	GenoLeWM	Carbon 0-shot	Δ
ClinVar coding	Accuracy $\uparrow$	0.7500	0.6875	+0.0625
	AUROC $\uparrow$	0.7344	0.9219	−0.1875
	AP $\uparrow$	0.8530	0.9519	−0.0989
	Balanced acc. $\uparrow$	0.7500	0.6875	+0.0625
ClinVar non-coding	Accuracy $\uparrow$	0.4375	0.6875	−0.2500
	AUROC $\uparrow$	0.5625	0.8750	−0.3125
	AP $\uparrow$	0.6055	0.9144	−0.3090
	Balanced acc. $\uparrow$	0.4375	0.6875	−0.2500
BRCA2 saturation	Spearman ρ $\uparrow$	0.1492	0.4769	−0.3277
TraitGym Mendelian	Spearman ρ $\uparrow$	−0.0280	−0.0839	+0.0559

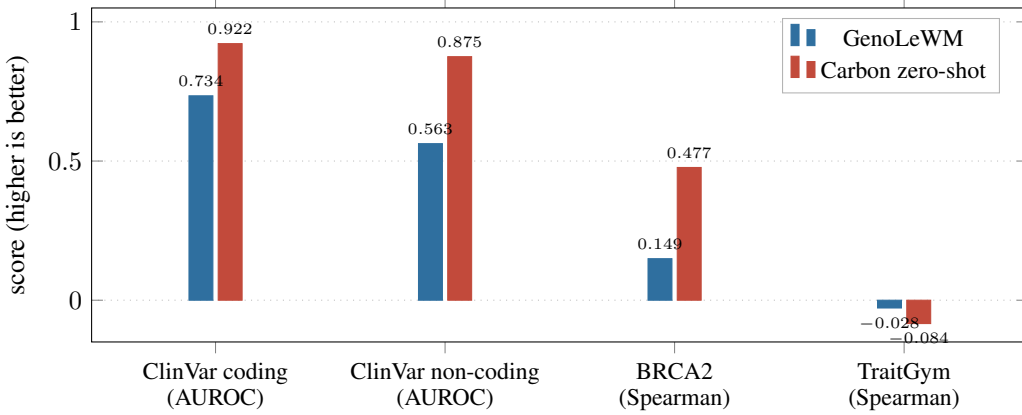


Figure 2: **Historical v0.2.1 VEP rows; withdrawn as a method comparison.** Carbon values are recovered as $\text{GenoLeWM} - \Delta$. The GenoLeWM ranking score came from an L_2 residual between unit-norm predictions and raw target states, so bar differences do not establish better or worse performance for the intended normalized-state method. The small slices add a separate power limitation; see §7.

bars. Because the corresponding scoring and rollout quality is invalid, the release supports only implementation-level timing and call-count observations, not an efficiency–quality or deployment claim.

5.4 Autoregressive rollout speed

KV-caching the action projections speeds autoregressive rollout by $2.41\times$ at $K=5$ (meeting its local $2\times$ target) and $2.47\times$ at $K=20$ (missing the $5\times$ target; fig. 5). The $K=20$ miss remains open and was explicitly rescope rather than closed. The published micro-benchmark used synthetic tensors on the H200 with batch 8, $d_{\text{state}}=1024$, $d_{\text{action}}=64$, CUDA, and bf16, but a benchmark-specific predictor shape ($d_{\text{hidden}}=1024$, 6 + 1 blocks, FFN 4096) rather than the released checkpoint’s exact architecture. It measures the cache implementation, not full-checkpoint or end-to-end rollout latency.

Table 4: **Historical rollout measurements and state-norm audit (v0.2.1-r1)**. Cosine values are scale-invariant historical measurements but are confounded by the invalid mixed-scale training objective, global-source/nominally-centered-target training mismatch, a one-hidden-token center offset, and train/rollout distribution shift. The L_2 columns are invalid comparisons: the predictor was unit norm while encoder source and target states were raw. No capability or mechanism is inferred. Recall@ k was 1.0 for both model and baseline and was non-discriminating.

Setting	Cosine to s_{t+1} $\uparrow$ (confounded)			L_2 to s_{t+1} (invalid)		
	GenoLeWM	Source-state	Δ	GenoLeWM	Source-state	Δ
Phased haplotypes	0.2889	0.9978	-0.7090	33.32	2.13	+31.19
Synthetic edit chains	0.3016	0.9912	-0.6896	28.80	3.17	+25.64

Setting	Mean $\ s_t\ _2$	Mean $\ s_{t+1}\ _2$	Mean $\ \hat{s}_{t+1}\ _2$
Phased haplotypes	33.982	33.595	1.000
Synthetic edit chains	29.253	29.089	1.000

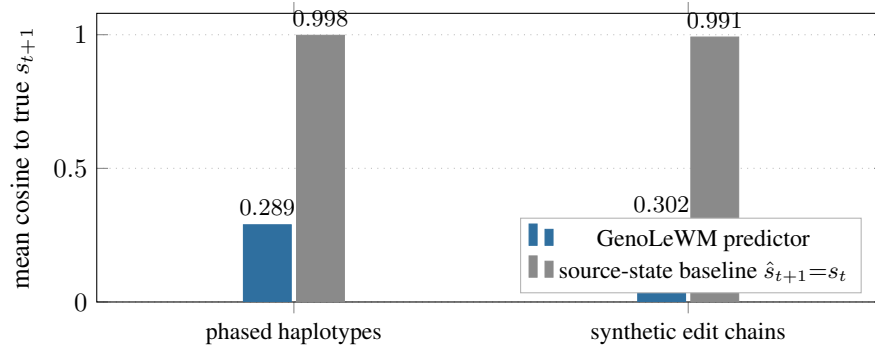


Figure 3: **Historical rollout cosine; mechanism withdrawn**. Prediction-to-target cosine is ≈ 0.29 – 0.30 and raw-source-to-target cosine is ≈ 0.99 . Cosine is scale-invariant, so these values remain historical measurements, but the predictor was trained with a mixed-scale L_2 term, mixed pooling coordinates, a one-hidden-token centering offset, and rollout differs from the training distribution. The gap therefore does not establish a latent-residual trap, action-conditioning failure, or any other mechanism (§6).

5.5 Planning

The released planning demonstration loads the checkpoint through the manifest-backed runtime and runs CEM against a synthetic multi-SNV target, terminating on patience after 384 evaluations with a historical best distance of 23.66 in ≈ 15.3 s. Because that distance compared incompatible state scales, it has no valid objective-space interpretation. The trace establishes only that the code path executed from released artifacts.

6 Discussion: Why the Released Result Is Not Interpretable

The original paper attributed the observed metrics to a structural latent-residual trap. That attribution is withdrawn. The current evidence establishes a contract violation and several confounders, not a mechanism.

Normalization contract violation (established). The encoder accepted `normalize: true` but emitted raw pooled states, whereas the predictor explicitly normalized its output. The roughly 29 – $34\times$ norm gap makes the released L_2 loss and metric dominated by operand scale. It also invalidates the previous identity-at-initialization equivalence, because normalizing the raw source is not equal to returning the raw source baseline.

Table 5: **Historical implementation timing (v0.2.1-r1, single NVIDIA H200).** The released single-variant score path includes process start, artifact loading, and two frozen Carbon-500M forward passes; its end-to-end latency is ≈ 115 s, without a component profile. The KV-cache micro-benchmark uses synthetic tensors at $d_{\text{state}}=1024$ on H200/CUDA bf16 with a benchmark-specific predictor shape. Because model-quality evaluation is invalidated, these rows do not support an efficiency–quality or deployment conclusion.

Quantity	Value
Single-variant latency (cold, 2 Carbon passes)	115,262.94 ms (≈ 115 s)
Batched throughput	0.3095 variants/s
Peak memory	1,966,149,632 B (≈ 1.83 GiB)
AR rollout speedup, $K=5$	$2.41\times$ (local $2\times$ target met)
AR rollout speedup, $K=20$	$2.47\times$ (RFC-0004 $5\times$ target <i>not</i> met, #42 open)

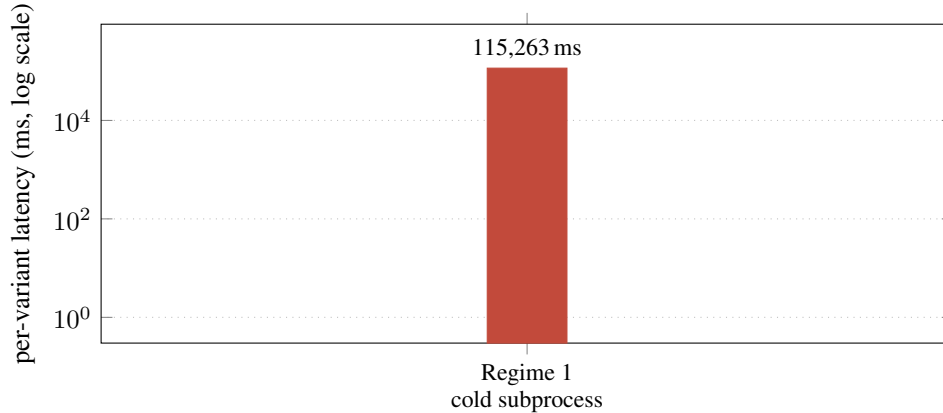


Figure 4: **Historical cold-process timing.** Only *Regime 1*, a cold geno-lewm-score subprocess (interpreter start, ~ 1 GiB Carbon load, two Carbon forward passes), is measured, at 115,262.94 ms. The release did not profile component timing, so this total cannot be apportioned between loading and compute. The “pay-Carbon-once” thesis applies to *Regime 2* (warm reference cache: one Carbon call on the edited window, in-process) and *Regime 3* (latent-only rollout/planning: zero further Carbon calls), both *unmeasured* on this checkpoint and therefore shown without bars. Because corresponding model quality is invalidated, this figure supports neither a deployment claim nor an efficiency–quality comparison.

Pooling-coordinate contract violation (established). Training source states used global-mean fallback, targets used nominally edit-centered pooling, and rollout evaluation supplied nominally edit-centered sources. Every nominal center omitted the leading `<dna>` token and was therefore one hidden token too far left; historical candidate states could also use different loci, while cache v1 did not commit a center token. This makes the latent operands coordinate-incompatible independently of normalization and sharpens the train/rollout mismatch; no metric from this lineage isolates a learned transition.

Transitive runtime-identity violation (established). The pinned Carbon custom tokenizer source performed an unpinned, network-capable Qwen/Qwen3-4B-Base load. A hash of that source file identifies the call site, not the mutable transitive tokenizer it resolves. Claims that the historical runtime was fully local or that the custom-tokenizer hash alone completed its identity are withdrawn. The corrected identity includes the local pure-DNA tokenizer implementation, but that code-path repair supplies no behavioral evidence.

Regularization mismatch (established). The phase2 label did not imply encoder adaptation. With no LoRA and a KL computed only on frozen targets, the regularizer had zero gradient with respect to all trainable parameters. The run therefore cannot test claims about Phase-2 joint representation learning or anti-collapse behavior.

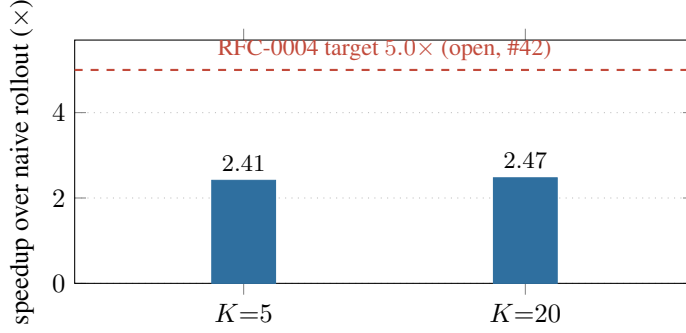


Figure 5: **Historical KV-cache micro-benchmark.** Cached ARPredictor rollout is $2.41\times$ ($K=5$) and $2.47\times$ ($K=20$) faster than naive repeated one-step prediction. The gain is real but short of the $5\times$ target at $K=20$, which remains open and was explicitly rescoped. Measured on synthetic tensors with batch 8, $d_{\text{state}}=1024$, $d_{\text{action}}=64$, CUDA, and bf16 on H200. The benchmark-specific predictor ($d_{\text{hidden}}=1024$, 6 + 1 blocks, FFN 4096) is not the exact released checkpoint.

Cosine and distribution shift (unresolved). Cosine removes scale at evaluation time, so the historical cosine values are not arithmetically invalid for that reason alone. They are nevertheless downstream of training with the mixed-scale L_2 term, and the rollout settings do not match the training distribution. Without a corrected, distribution-matched evaluation, the cosine gap cannot identify whether the cause is normalization, optimization, distribution shift, action-conditioning failure, or encoder geometry.

Mechanistic attribution withdrawn. The latent-residual trap, edit-agnostic encoder geometry, and identity attraction remain plausible hypotheses, but this release does not distinguish them. We therefore withdraw the claims that freezing caused the observed gap, that the model learned a distorted copy, or that edit-conditioned prediction is unlearnable in this regime. Those questions reopen only after the corrected control run.

7 Limitations

The primary limitation is a validity failure, not merely low power. (i) *State contract*: `normalize: true` was ignored for source and target states, while predictions were unit norm; all L_2 -dependent results are invalid for the intended method. (ii) *Pooling coordinates*: training sources were global, targets and rollout sources used a one-token-left “center,” historical candidates could use different loci, and cache v1 omitted that identity field. (iii) *Runtime identity*: the pinned custom tokenizer performed an unpinned, network-capable transitive Qwen load. (iv) *Training semantics*: the run was labeled phase2 but used no LoRA, and its target-only KL supplied no gradient. (v) *Distribution shift*: historical rollout cosine is confounded by mismatch between training and rollout distributions. (vi) *Statistical power*: ClinVar slices are $N=16$ and rollout slices are $N=8$. (vii) *Coverage and configuration drift*: training and holdout coverage are narrow, while pooling radius and predictor depth diverge from design RFCs. (viii) *Efficiency and planning*: warm-cache latency was not measured, the AR speed test used synthetic tensors and a benchmark-specific predictor shape, and planning distances are invalid under the state mismatch. (ix) *Receipts*: checksums certify included identity, not transitive dependencies, execution semantics, scientific validity, or privacy. These limitations preclude model-quality and mechanistic conclusions from this release.

8 Historical Released Measurements

For auditability, we preserve the released rows without labeling them scientific findings. Historical VEP deltas include ClinVar non-coding AUROC -0.3125 , BRCA2 Spearman -0.328 , and ClinVar coding AUROC -0.1875 ; they derive from an invalid L_2 surprise score. Historical rollout cosine deltas are -0.709 and -0.690 ; they are scale-invariant but confounded. Historical rollout L_2 deltas are $+31.19$ and $+25.64$; they are invalid because the operands had incompatible norms.

The synthetic-tensor $K=20$ AR micro-benchmark measured $2.47\times$, the cold subprocess latency was about 115 s, and the planning trace stopped on patience at historical distance 23.66. The correction changes the interpretation, not the archived bytes.

9 Reproducibility

Every historical value is bound to the included identities in table 6; the unpinned transitive tokenizer remains outside that graph. This enables forensic inspection of the flawed run, not exact self-contained reproduction of its runtime or reproduction of the intended method. The corrected experiment must not replace files beneath the existing identity or present a patched evaluator against the old checkpoint as a corrected result. It must issue a new model id, dataset/evaluation manifest, configuration hash, and receipt graph; resolve tokenization exclusively from committed local files; verify the DNA-content offset; record observed state norms at train and evaluation time; prove that `normalize: true` changes the tensors consumed by the predictor, loss, cache, scorer, rollout, and planner; and verify nonzero gradients for every regularizer described as active. Until that evidence exists, `v0.2.1-r1` remains an archived invalidated run.

10 Future Work

The first future experiment is a correction control, not a new modeling direction. It must (1) tokenize from a self-contained committed local package and validate the DNA control-token layout; (2) apply the same explicit normalization view to live and cached source and target states; (3) use the token-layout-derived shared pooling center for every compared state and commit that center in cache identity; (4) assert unit norms before loss, scoring, rollout, and planning; (5) train a true frozen-encoder baseline under `phase1`; (6) if Phase 2 is tested, add actual LoRA parameters and verify a nonzero KL gradient into them; (7) match train and rollout distributions or pre-register the shift and corresponding stratified analysis; (8) include shuffled-action and source-state controls; (9) use adequately powered evaluation slices and repeated seeds; and (10) publish under a new run identity. Only after this control may hypotheses such as residual-centric objectives, edit-direction probes, LoRA adaptation, alternative encoders, or scaling laws be evaluated. The old mechanistic program is retained as a set of hypotheses, not as consequences of `v0.2.1-r1`.

11 Conclusion

We built and released an action-conditioned genomic latent-prediction pipeline, but the `v0.2.1-r1` experiment did not implement its declared state contract. Raw encoder states with norms near 29–34 were compared against unit-norm predictions; source, target, and candidate pooling coordinates were inconsistent and nominal centers were one hidden token left; the tokenizer could resolve an unpinned network dependency; the run was labeled Phase 2 without LoRA; and its target-only KL had no gradient. We therefore withdraw the released L_2 -based results, all VEP and planning interpretations derived from them, the mechanistic latent-residual attribution, historical self-contained-runtime claims, and all model-capability conclusions. Historical numbers remain attached to their included identities for auditability. The scientific question remains open and requires a corrected, self-contained, coordinate-matched, distribution-aware, adequately powered rerun under a new content-addressed identity.

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A Notation

Symbol	Meaning
$w_{\text{ref}}, w_{\text{alt}}$	reference / edited DNA window (4,096 bp released; 12,288 bp code default)
$\text{enc}(\cdot)$	Carbon-500M layer-20, centered-mean pool ($r=8$); normalization is a separate contract
$s_t = \text{enc}(w_{\text{ref}})$	reference state; raw/global in training, raw/centered in rollout
$s_{t+1} = \text{enc}(w_{\text{alt}})$	post-edit target; raw/centered in the release
$\hat{s}_{t+1} = g(s_t, a)$	predicted post-edit state; unit norm in the release (eq. (2))
$\Delta = s_{t+1} - s_t$	edit residual; released magnitude is not comparable with $\hat{s}_{t+1}$
a, a_{emb}	edit action and its embedding, $\mathbb{R}^{64}$
g	cross-attention residual predictor (6 cross + 2 self, $d_{\text{hidden}}=768$)
α, β	prediction-loss weights (1.0, 0.1); released $\gamma=0.5$ KL had no gradient
$\sigma_{\text{raw}}, \sigma_{\text{cal}}$	raw / context-calibrated surprise (165 buckets)
$\text{score}_{\text{Carbon}}$	$\log P(w_{\text{ref}}) - \log P(w_{\text{alt}})$ (zero-shot baseline)
N	evaluated variants per slice: 16/16/32/32/8/8

B Artifact Identity

Table 6: **Content-addressed identity of the invalidated release.** Every historical number is bound to these identities through checksum receipts, and all 28 suite outputs verified at the byte and graph level. The receipts did not verify normalization semantics, token-layout offsets, transitive tokenizer dependencies, or gradient paths. A corrected run must use a new identity (§9).

Field	Value
Model release	geno-lewm-v0.2.1-r1
Model id	sha256:cddb8f3b9671...f5f3ed6af4
Dataset snapshot	geno-lewm-data-v0.2.1-r1
Commit	d9b06815cf8e...5aa4154d
Hardware	NVIDIA H200 (Linux x86_64, glibc 2.35)
Suite outputs	28 / 28 verified (release_inputs: pass)

C Citation Verification Notes

Bibliography metadata was checked against publisher pages, DOIs, or arXiv where available. One source caveat applies: the Carbon-500M entry uses an organizational author because the model card exposes no formal author list or technical-report citation. The BRCA2 score bytes used by the released benchmark are from MaveDB score set urn:mavedb:00001242-a-1, whose primary-publication record identifies Sahu et al. [18]; the similarly named Findlay et al. saturation-genome-editing study concerns BRCA1 and is not this source. A small number of page numbers in otherwise-verified entries are inferred and flagged in the bibliography file.